

西方经济学

Part 2 Classical Theory

THE ECONOMY IN THE LONG RUN

Lecture 2C A Classical Open Economy

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(1) M14.1; S11.1, S11.2. ¹

(2) 其他资料: [GM]

- THE OPEN ECONOMY



¹M 指代马工程教材, S 指代课外阅读材料沈坤荣教程。

学习目标

西方经济学

- (1) 掌握小国经济模型。
- (2) 掌握大国经济模型。
- (3) 领会马工程教材精神。



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Outline

1 Exchange Rates

- Exchange Rates
- The Law of One Price
- Purchasing Power Parity
- Net Exports
- Uncovered Interest Parity
- Assumptions

2 Case: Small Home vs. Large Foreign

3 Case: Large Home vs. Large Foreign

4 马工程教材疑难重点

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Exchange Rates

Definition 1

The *nominal exchange rate*, denoted by e , is the price of the domestic currency in terms of foreign currency. The *real exchange rate*, denoted by ϵ , is the price of domestic goods in terms of goods of a foreign country. ϵ can be written as $\epsilon = e \times P/P^*$, where P denotes the price of domestic goods and P^* the price of goods of a foreign country.

马工程观点 (M, 2019, p.182)

名义汇率是两个国家（或地区）货币的相对价格，即一种货币能兑换另一种货币的数量，用 e 表示。名义汇率有两种不同的标价方法。一种是直接标价法，是用本国货币形式表示的国外货币的价格，即购买 1 单位或 100 单位的外币应该付出多少本国货币，故又称为应付标价法。这样，当本币升值时，购买单位外币所必须支付的本币就减少，即本币升值，汇率下降，也就是说，汇率升降与本国货币对外价值的高低成反比。另一种为间接标价法，即用国外货币表示本国货币的价格。它以购买一定单位（如 1 单位）的本国货币为标准来计算应收多少单位外币，故又称为应收标价法。由于这种标价法将本国货币设定为一定的数额，当本币币值上升时，单位本币所能兑换的外币就增加，也就是说，外币价值的高低和汇率的升降成反比。为了方便起见，本章采用间接标价法。

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Proposition 2 (The Law of One Price)

If markets are perfectly competitive and there are no frictions, then identical goods sold in different locations must sell for the same price in terms of a common currency.

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Purchasing Power Parity

Proposition 3 (Purchasing Power of Parity [PPP])

*If markets are perfectly competitive and there are no frictions, then the real exchange rate ϵ is equal to 1.*¹

Proposition 3 implies $e = P^*/P$, which is called absolute PPP. In form of growth rate, $\frac{\dot{e}}{e} = \frac{\dot{P}^*}{P^*} - \frac{\dot{P}}{P}$, which is called relative PPP.

¹Parity: the quality or state of being equal or equivalent; equivalence of a commodity price expressed in one currency to its price expressed in another. — [Merriam-Webster](#)

Net Exports

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If ϵ is high, foreign goods are relatively cheap and domestic goods are relatively expensive. Other things being equal, a higher real exchange rate implies that a country imports more and exports less. Therefore the net exports NX is decreasing in ϵ .

$$NX = NX(\epsilon), \quad NX'(\epsilon) < 0$$



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Uncovered Interest Parity

Proposition 4

If capital mobility is perfect, then equilibrium in the foreign exchange market implies the uncovered interest parity:

$$1 + i_t = (1 + i_t^*) \frac{e_t}{\mathbb{E}_t e_{t+1}}$$

or the uncovered real interest parity:

$$1 + r_t = (1 + r_t^*) \frac{\epsilon_t}{\mathbb{E}_t \epsilon_{t+1}}$$



Assumptions

Definition 5

An economy is a *small open economy* if it is a small part of the world market and the net capital flow of the economy has only a negligible effect on the world interest rate. An economy is a *large open economy* if it is a large part of the world market and the net capital flow of the economy has a significant effect on the world interest rate.

Assumption 1 (Perfect Capital Mobility [PCM])

Residents of a country have full access to world financial markets.

Assumption 2 (Static Expectations of ϵ)

$$\mathbb{E}_t \epsilon_{t+1} = \epsilon_t.$$

Assumption 3

PPP does not always hold.

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Case: Small Home vs. Large Foreign (Assumptions 1–3)

Assumptions 1 and 2 imply $r = r^*$. For a small open economy, r^* is exogenous.

1. Market for Goods and Services

$$\bar{Y} = C(\bar{Y} - \bar{T}) + I(r^*) + \bar{G} + NX(\epsilon).$$

The only endogenous variable is $NX(\epsilon)$ which plays a role of equating the left-hand side and the right-hand side.

2. Market for Loanable Funds

$$\bar{S} = I(r^*) + NX(\epsilon).$$

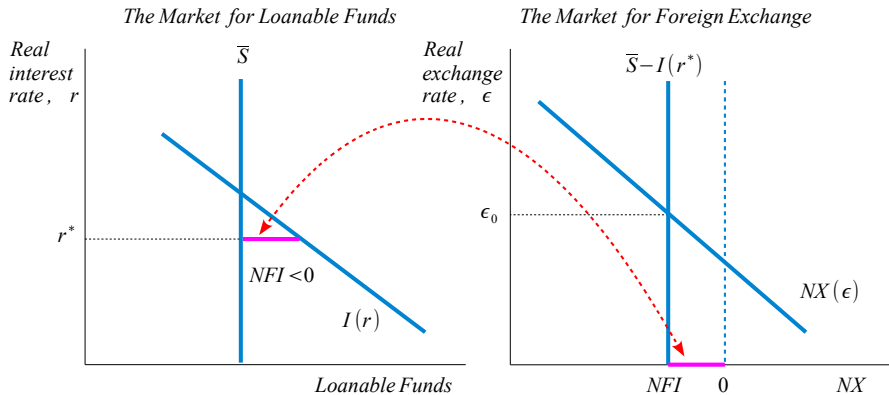
The difference between the supply and demand for loanable funds is called net foreign investment (NFI).

3. Market for Foreign Exchange

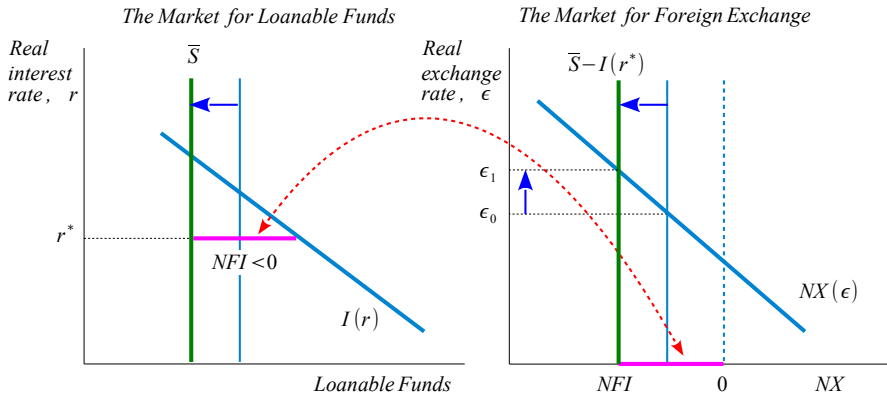
$$\bar{S} - I(r^*) = NX(\epsilon).$$

$\bar{S} - I(r^*)$ captures supply of domestic currency or demand for foreign currency; NX captures demand for domestic currency or supply of foreign currency.

The Complete Model for the Classical Small Open Economy with PCM and Floating ϵ



A Reduction in Home's Saving



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Case: Large Home vs. Large Foreign (Assumptions 1–3)

Large Home :
$$NX(\epsilon) = \left[\bar{Y} - C(\bar{Y} - \bar{T}) - G \right] - I(r);$$

Large Foreign :
$$NX^*(1/\epsilon) = \left[\bar{Y}^* - C^*(\bar{Y}^* - \bar{T}^*) - G^* \right] - I^*(r^*);$$

Perfect Capital Mobility :
$$r = r^*;$$

The World is closed :
$$NX(\epsilon) + NX^*(1/\epsilon) \equiv 0.$$

It follows that

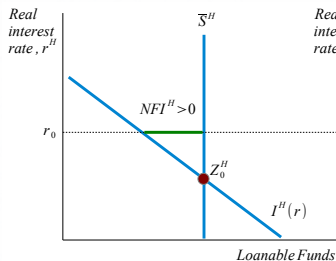
$$\bar{S} + \bar{S}^* = I(r) + I^*(r);$$

$$NX(\epsilon) = \bar{S} - I(r).$$

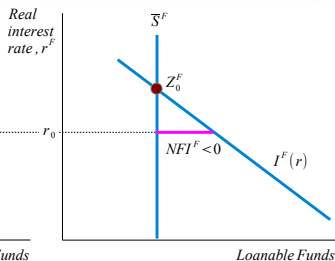


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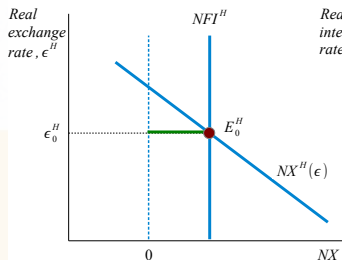
Case: Large Home vs. Large Foreign (Assumptions 1–3)



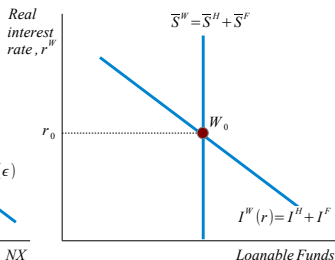
(a) Home Country



(b) Foreign Country

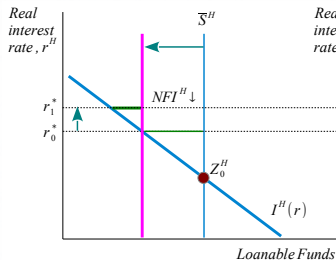


(c) Foreign Exchange Market

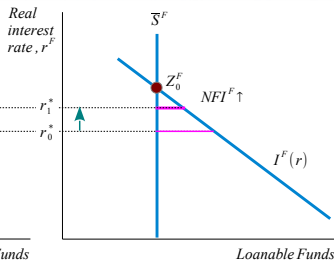


(d) The World

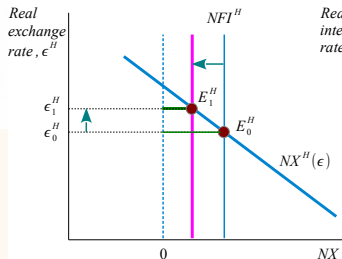
A Reduction in Home's Saving



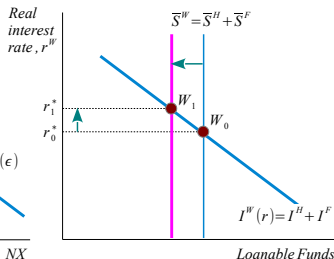
(a) Home Country



(b) Foreign Country



(c) Foreign Exchange Market



(d) The World

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疑难重点

西方经济学

- (1) 理解实际汇率的计算。
- (2) 掌握 The law of one price, PPP, uncovered interest parity.
- (3) 在 Small home-large foreign 模型中, $Y, C, I, S, NX, r, \epsilon$ 如何决定?
- (4) 在 Large home-large foreign 模型中, $Y, C, I, S, NX, r, \epsilon$ 如何决定?

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1 (E2, p.39)

根据马工程教材观点，为什么 1830 年后的经济学被马克思界定为“庸俗经济学”？

2 (E2, p.3)

根据马工程教材观点，应当如何评析西方经济学的科学因素和阶级属性？



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